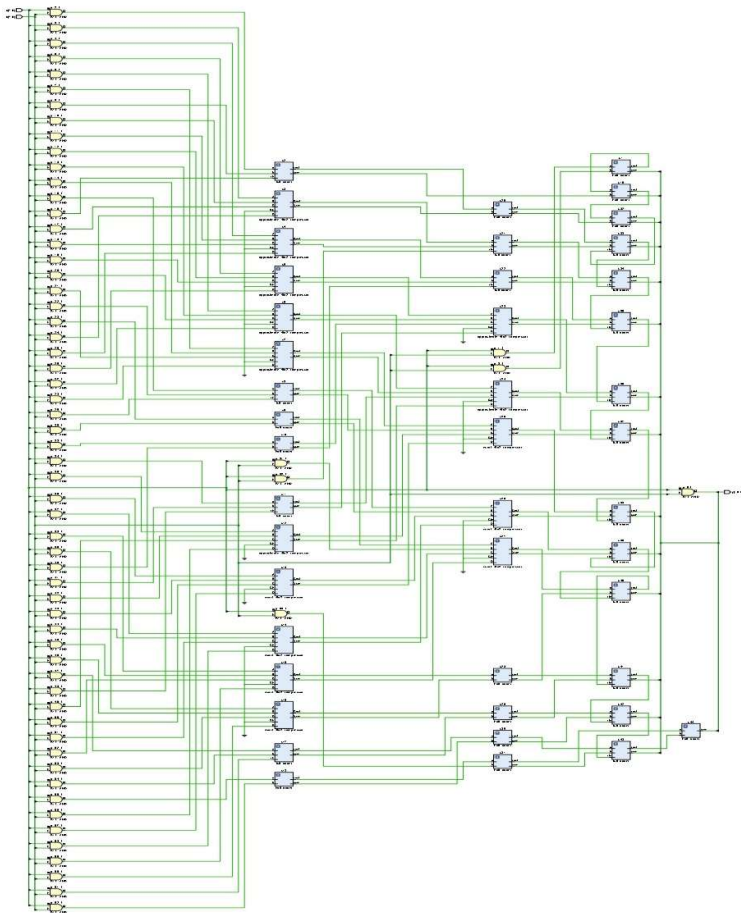


Results of Base Paper Dadda:

Simulation:



Schematic:



Tcl Console

Messages

Log

Reports

Design Runs

Timing

Power

DRC

Package Pins

I/O Ports

?

⌵

⌵

Q

⌵

⌵

⌵

⌵

Summary

Settings

Summary (10.561 W, Margin: 99%)

Power Supply

Utilization Details

Hierarchical (10.45 W)

Signals (0.641 W)

Data (0.641 W)

Logic (0.459 W)

I/O (9.35 W)

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power:

10.561 W

Design Power Budget:

Not Specified

Process:

typical

Power Budget Margin:

NA

Junction Temperature:

44.9C

Thermal Margin:

40.1°C (21.1 W)

Ambient Temperature:

25.0 °C

Effective ΘJA:

1.9°C/W

Power supplied to off-chip devices:

0 W

Confidence level:

Low

Launch Power Constraint Advisor

to find and fix invalid switching activity

On-Chip Power

99%

Dynamic: 10.450 W (99%)

90%

Signals: 0.641 W (6%)

Logic: 0.459 W (4%)

I/O: 9.350 W (90%)

Device Static: 0.111 W (1%)

impl_1 (saved)

[illegible]

Tcl ConsoleMessagesLogReportsDesign RunsDRCPowerTiming

Unconstrained Paths - NONE - NONE - Setup

General Information

Timer Settings

Design Timing Summary

Methodology Summary

Check Timing (0)

Intra-Clock Paths

Inter-Clock Paths

Other Path Groups

User Ignored Paths

Unconstrained Paths

NONE to NONE

Setup (10)

Hold (10)

Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception
Path 1	∞	9	8	20	a[6]	y[15]	14.480	4.873	9.607	∞	input port clock		
Path 2	∞	9	8	20	a[6]	y[14]	14.024	4.631	9.393	∞	input port clock		
Path 3	∞	8	7	20	a[6]	y[12]	12.917	4.715	8.203	∞	input port clock		
Path 4	∞	8	7	20	a[6]	y[13]	12.784	4.514	8.270	∞	input port clock		
Path 5	∞	7	6	20	a[6]	y[11]	12.378	4.602	7.777	∞	input port clock		
Path 6	∞	7	6	20	a[6]	y[10]	12.358	4.418	7.940	∞	input port clock		
Path 7	∞	6	5	20	a[6]	y[9]	11.602	4.300	7.303	∞	input port clock		
Path 8	∞	6	5	20	a[6]	y[8]	11.443	4.491	6.952	∞	input port clock		
Path 9	∞	6	5	20	a[6]	y[7]	11.009	4.260	6.750	∞	input port clock		
Path 10	∞	6	5	15	a[3]	y[6]	10.798	4.268	6.530	∞	input port clock		

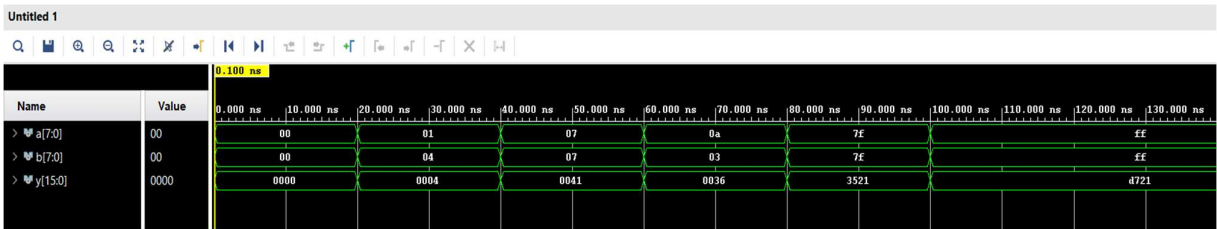
Timing Summary - timing.1

Area:

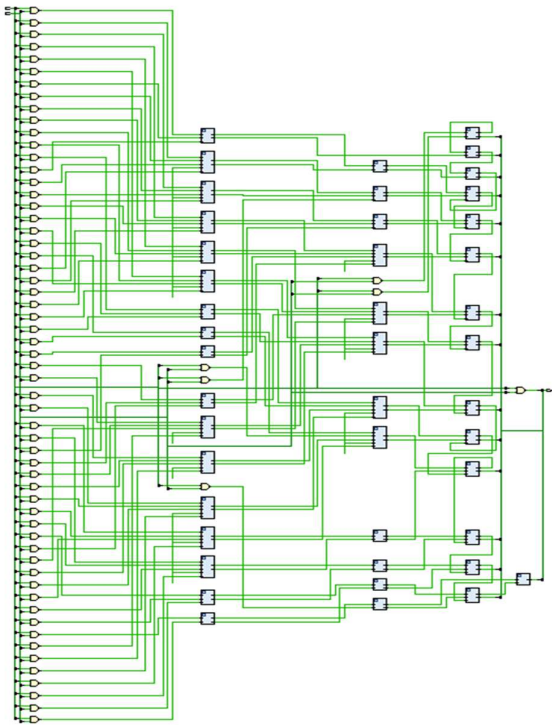
Site Type	Used	Available	Util%
Slice LUTs*	71	20800	0.34
LUT as Logic	71	20800	0.34
Bonded IOB	32	106	30.19

Results of Proposed Dadda:

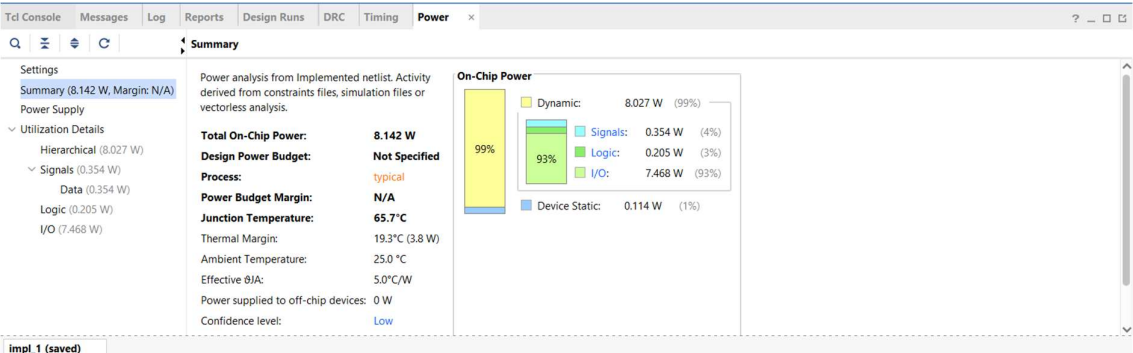
Simulation:



Schematic:



Power:



Timing:

Unconstrained Paths - NONE - NONE - Hold														
	Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock U
General Information	Path 11	∞	3	12	b[5]	y[5]	2.252	1.311	0.940	∞	input port clock			
Timer Settings	Path 12	∞	3	14	b[6]	y[8]	2.259	1.314	0.945	∞	input port clock			
Design Timing Summary	Path 13	∞	3	12	b[5]	y[6]	2.270	1.325	0.945	∞	input port clock			
Methodology Summary	Path 14	∞	3	11	b[4]	y[7]	2.278	1.327	0.951	∞	input port clock			
Check Timing (0)	Path 15	∞	3	14	b[6]	y[15]	2.305	1.324	0.981	∞	input port clock			
Intra-Clock Paths	Path 16	∞	3	11	a[1]	y[2]	2.336	1.324	1.012	∞	input port clock			
Inter-Clock Paths	Path 17	∞	3	9	b[3]	y[10]	2.337	1.353	0.984	∞	input port clock			
Other Path Groups	Path 18	∞	3	11	a[1]	y[1]	2.355	1.321	1.034	∞	input port clock			
User Ignored Paths	Path 19	∞	3	11	b[4]	y[4]	2.358	1.400	0.958	∞	input port clock			
Unconstrained Paths	Path 20	∞	3	14	b[6]	y[14]	2.381	1.383	0.998	∞	input port clock			
NONE to NONE														
Setup (10)														
Hold (10)														

Unconstrained Paths - NONE - NONE - Setup														
	Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock U
General Information	Path 1	∞	6	12	b[7]	y[14]	10.379	4.254	6.124	∞	input port clock			
Timer Settings	Path 2	∞	6	12	b[7]	y[15]	10.139	4.030	6.109	∞	input port clock			
Design Timing Summary	Path 3	∞	6	13	a[0]	y[5]	10.105	4.026	6.079	∞	input port clock			
Methodology Summary	Path 4	∞	6	13	a[0]	y[6]	9.977	4.039	5.938	∞	input port clock			
Check Timing (0)	Path 5	∞	5	13	a[0]	y[4]	9.643	4.128	5.515	∞	input port clock			
Intra-Clock Paths	Path 6	∞	5	11	a[1]	y[9]	9.476	3.933	5.543	∞	input port clock			
Inter-Clock Paths	Path 7	∞	6	8	a[4]	y[12]	9.339	4.272	5.068	∞	input port clock			
Other Path Groups	Path 8	∞	5	8	a[4]	y[11]	9.307	4.362	4.946	∞	input port clock			
User Ignored Paths	Path 9	∞	4	11	a[1]	y[10]	9.206	3.803	5.403	∞	input port clock			
Unconstrained Paths	Path 10	∞	5	8	a[3]	y[7]	9.076	4.127	4.949	∞	input port clock			
NONE to NONE														
Setup (10)														
Hold (10)														

Area:

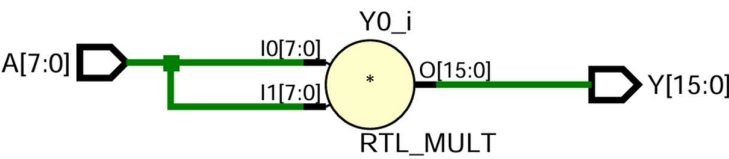
Site Type	Used	Available	Util%
Slice LUTs*	41	20800	0.20
LUT as Logic	41	20800	0.20
Bonded IOB	32	106	30.19

Results of Proposed Squarer:

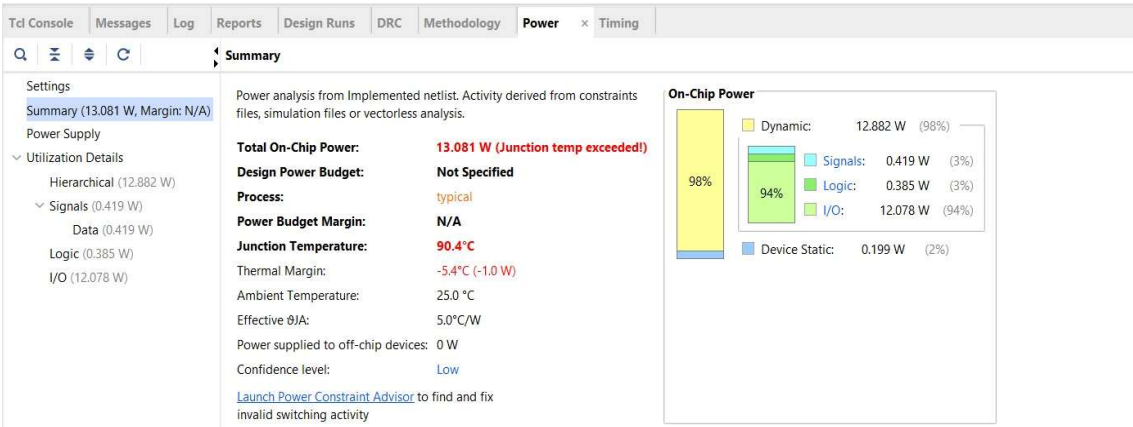
Simulation:



Schematic:



Power:



Timing:

Timing Summary - Impl_1 (saved)													
Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Un
Path 11	∞	2	20	A[0]	Y[0]	1.757	1.296	0.461	-∞	input port clock			
Path 12	∞	4	20	A[0]	Y[2]	2.601	1.466	1.135	-∞	input port clock			
Path 13	∞	5	23	A[3]	Y[3]	2.606	1.504	1.102	-∞	input port clock			
Path 14	∞	6	23	A[3]	Y[5]	2.726	1.615	1.111	-∞	input port clock			
Path 15	∞	6	26	A[6]	Y[6]	2.741	1.636	1.105	-∞	input port clock			
Path 16	∞	6	25	A[4]	Y[11]	2.748	1.643	1.105	-∞	input port clock			
Path 17	∞	6	22	A[2]	Y[7]	2.769	1.632	1.137	-∞	input port clock			
Path 18	∞	6	23	A[3]	Y[4]	2.777	1.640	1.136	-∞	input port clock			
Path 19	∞	7	26	A[6]	Y[8]	2.809	1.737	1.072	-∞	input port clock			
Path 20	∞	6	24	A[7]	Y[9]	2.836	1.630	1.206	-∞	input port clock			

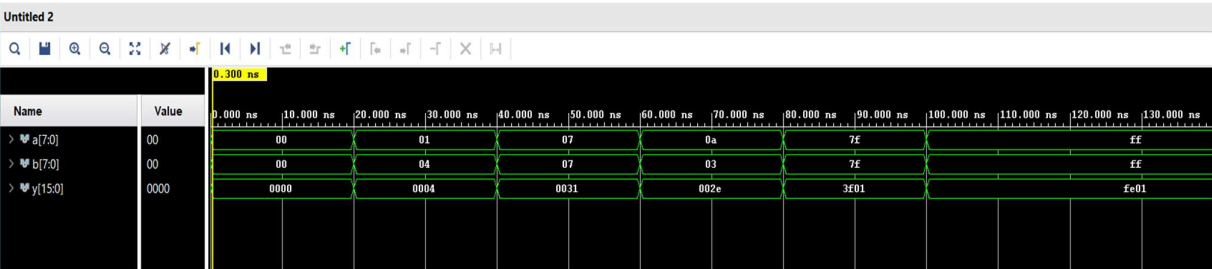
Timing Summary - impl_1 (saved)													
General Information	Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception
Timer Settings	Path 1	∞	9	20	A[0]	Y[15]	12.827	6.084	6.743	∞	input port clock		
Design Timing Summary	Path 2	∞	9	20	A[0]	Y[13]	12.798	6.106	6.692	∞	input port clock		
Methodology Summary (12)	Path 3	∞	9	20	A[0]	Y[14]	12.714	6.023	6.691	∞	input port clock		
> Check Timing (0)	Path 4	∞	9	20	A[0]	Y[12]	12.676	5.990	6.685	∞	input port clock		
Intra-Clock Paths	Path 5	∞	8	20	A[0]	Y[10]	12.499	5.809	6.690	∞	input port clock		
Inter-Clock Paths	Path 6	∞	8	20	A[0]	Y[11]	12.425	5.873	6.551	∞	input port clock		
Other Path Groups	Path 7	∞	8	20	A[0]	Y[9]	12.139	5.454	6.685	∞	input port clock		
User Ignored Paths	Path 8	∞	8	20	A[0]	Y[8]	11.485	5.112	6.373	∞	input port clock		
Unconstrained Paths	Path 9	∞	7	27	A[1]	Y[7]	11.424	5.435	5.988	∞	input port clock		
NONE to NONE	Path 10	∞	8	27	A[1]	Y[6]	11.196	5.363	5.833	∞	input port clock		
Setup (10)													
Hold (10)													

Area:

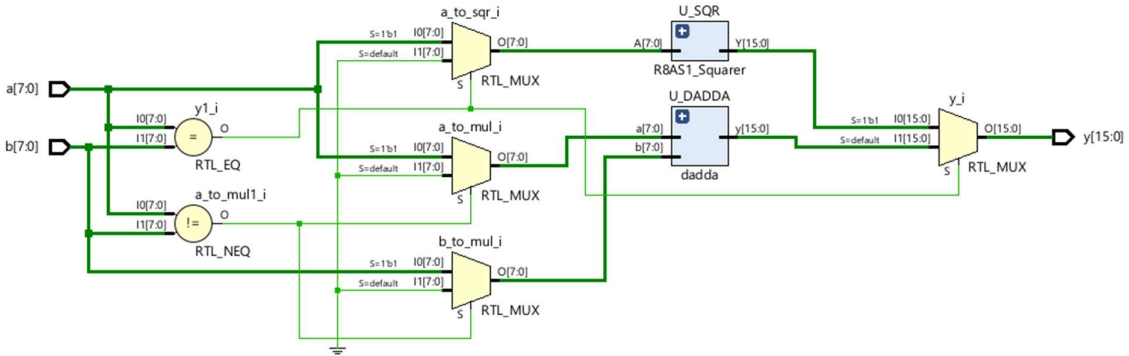
Site Type	Used	Available	Util%
Slice LUTs*	46	20800	0.22
LUT as Logic	46	20800	0.22
Bonded IOB	24	106	22.64

Results of Integrated Base Paper Dadda and Squarer:

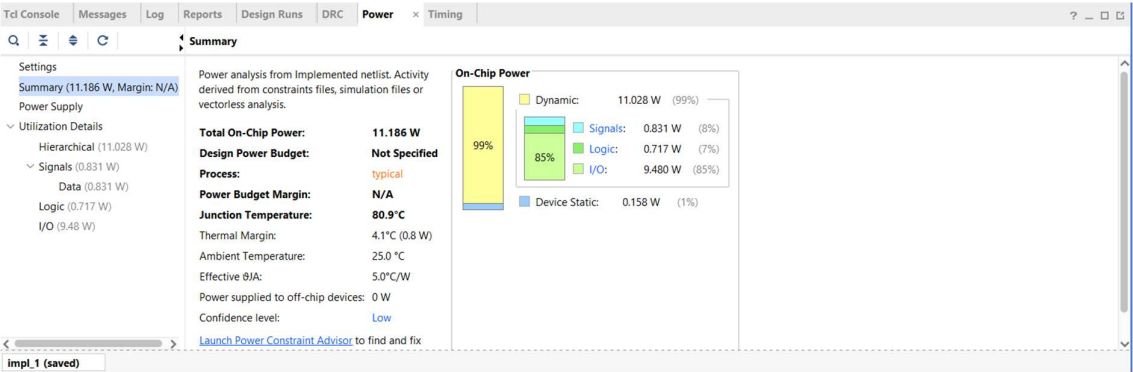
Simulation:



Schematic:



Power:



Timing:

The Timing summary window displays the following information:

- General Information:** Unconstrained Paths - NONE - NONE - Hold
- Timer Settings:** Path 11, Path 12, Path 13, Path 14, Path 15, Path 16, Path 17, Path 18, Path 19, Path 20
- Methodology Summary:** Check Timing (0)
- Intra-Clock Paths:** Path 11, Path 12, Path 13, Path 14, Path 15, Path 16, Path 17, Path 18, Path 19, Path 20
- Inter-Clock Paths:** Path 11, Path 12, Path 13, Path 14, Path 15, Path 16, Path 17, Path 18, Path 19, Path 20
- Other Path Groups:** User Ignored Paths
- User Ignored Paths:** Unconstrained Paths
- Unconstrained Paths:** NONE to NONE
- Setup (10):** Hold (10)

Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Un
Path 11	∞	3	22	b[2]	y[2]	2.192	1.341	0.850	∞	input port clock			
Path 12	∞	3	43	a[0]	y[1]	2.354	1.327	1.028	∞	input port clock			
Path 13	∞	3	43	a[0]	y[0]	2.426	1.386	1.041	∞	input port clock			
Path 14	∞	5	20	b[6]	y[7]	2.504	1.408	1.095	∞	input port clock			
Path 15	∞	4	22	b[2]	y[3]	2.515	1.387	1.128	∞	input port clock			
Path 16	∞	4	20	b[6]	y[13]	2.544	1.370	1.174	∞	input port clock			
Path 17	∞	4	21	b[5]	y[12]	2.547	1.363	1.184	∞	input port clock			
Path 18	∞	4	15	b[7]	y[14]	2.619	1.365	1.255	∞	input port clock			
Path 19	∞	4	15	b[7]	y[15]	2.664	1.371	1.293	∞	input port clock			
Path 20	∞	4	15	b[7]	y[10]	2.680	1.399	1.282	∞	input port clock			

The Timing summary window displays the following information:

- General Information:** Unconstrained Paths - NONE - NONE - Setup
- Timer Settings:** Path 1, Path 2, Path 3, Path 4, Path 5, Path 6, Path 7, Path 8, Path 9, Path 10
- Methodology Summary:** Check Timing (0)
- Intra-Clock Paths:** Path 1, Path 2, Path 3, Path 4, Path 5, Path 6, Path 7, Path 8, Path 9, Path 10
- Inter-Clock Paths:** Path 1, Path 2, Path 3, Path 4, Path 5, Path 6, Path 7, Path 8, Path 9, Path 10
- Other Path Groups:** User Ignored Paths
- User Ignored Paths:** Unconstrained Paths
- Unconstrained Paths:** NONE to NONE
- Setup (10):** Hold (10)

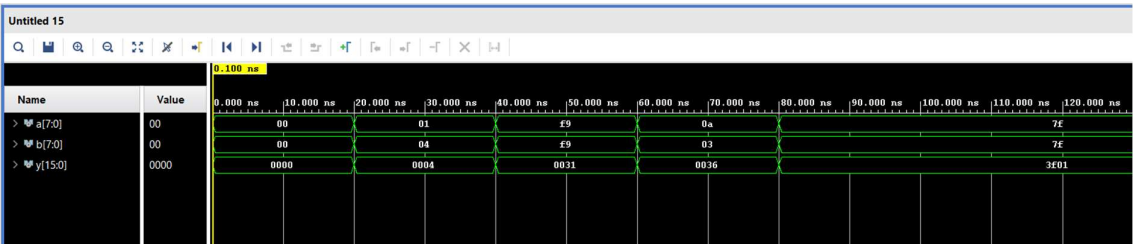
Name	Slack	Levels	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception	Clock Un
Path 1	∞	14	133	a[3]	y[12]	17.805	6.767	11.039	∞	input port clock			
Path 2	∞	14	133	a[3]	y[13]	17.777	6.666	11.111	∞	input port clock			
Path 3	∞	14	133	a[3]	y[11]	17.551	6.662	10.889	∞	input port clock			
Path 4	∞	15	133	a[3]	y[15]	17.427	6.473	10.954	∞	input port clock			
Path 5	∞	13	133	a[3]	y[9]	17.372	6.491	10.882	∞	input port clock			
Path 6	∞	13	133	a[3]	y[10]	17.253	6.549	10.704	∞	input port clock			
Path 7	∞	11	42	a[7]	y[14]	17.210	4.871	12.339	∞	input port clock			
Path 8	∞	13	133	a[3]	y[8]	16.949	6.094	10.855	∞	input port clock			
Path 9	∞	12	133	a[3]	y[7]	15.029	5.603	9.427	∞	input port clock			
Path 10	∞	11	133	a[3]	y[6]	14.202	5.252	8.950	∞	input port clock			

Area:

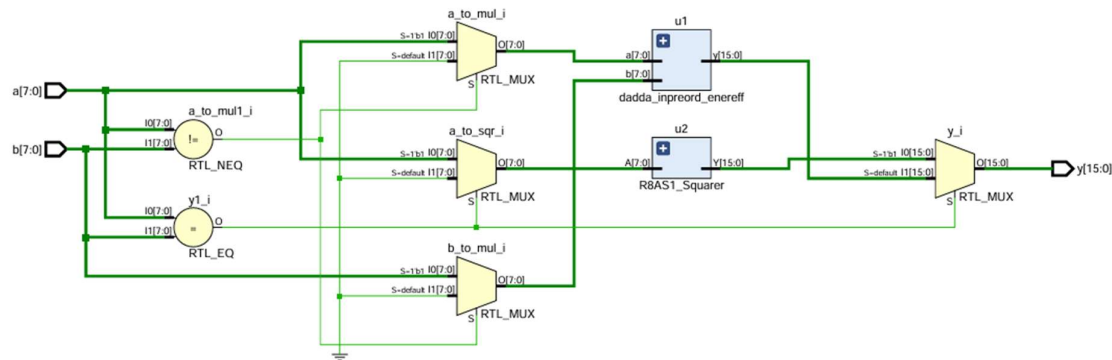
Site Type	Used	Available	Util%
Slice LUTs*	171	20800	0.82
LUT as Logic	171	20800	0.82
Bonded IOB	32	106	30.19

Results of Proposed Integrated Proposed Dadda and Squarer:

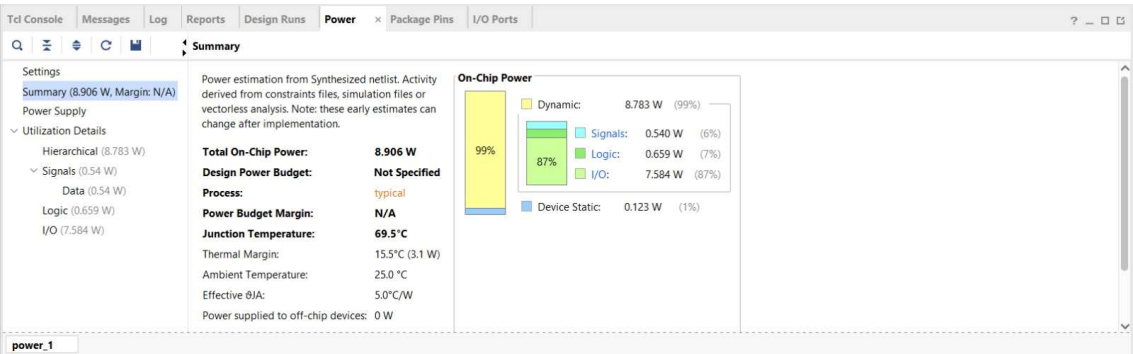
Simulation:



Schematic:



Power:



Timing:

Tcl ConsoleMessagesLogReportsDesign RunsDRCPowerTiming

Q

≡

⌕

●

Q

≡

⌕

●

Unconstrained Paths - NONE - NONE - Hold

?

⌕

□

□

□

General Information

Timer Settings

Design Timing Summary

Methodology Summary

Check Timing (0)

Intra-Clock Paths

Inter-Clock Paths

Other Path Groups

User Ignored Paths

Unconstrained Paths

NONE to NONE

Setup (10)

Hold (10)

Name

Slack

Levels

High Fanout

From

To

Total Delay

Logic Delay

Net Delay

Requirement

Source Clock

Destination Clock

Exception

Clock Un

Path 11

∞

3

12

b[1]

y[1]

2.316

1.324

0.992

∞

input port clock

Path 12

∞

3

13

b[7]

y[14]

2.340

1.320

1.020

∞

input port clock

Path 13

∞

4

12

b[2]

y[3]

2.357

1.387

0.970

∞

input port clock

Path 14

∞

3

13

b[4]

y[4]

2.395

1.331

1.063

∞

input port clock

Path 15

∞

4

12

b[2]

y[2]

2.395

1.386

1.008

∞

input port clock

Path 16

∞

4

15

b[5]

y[7]

2.454

1.352

1.102

∞

input port clock

Path 17

∞

3

13

b[7]

y[15]

2.477

1.326

1.152

∞

input port clock

Path 18

∞

4

15

b[5]

y[8]

2.518

1.348

1.170

∞

input port clock

Path 19

∞

4

13

b[4]

y[11]

2.622

1.394

1.228

∞

input port clock

Path 20

∞

4

15

b[5]

y[5]

2.639

1.421

1.218

∞

input port clock

Timing Summary - impl_1 (saved)

Tcl Console

Messages

Log

Reports

Design Runs

DRC

Power

Timing

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

?

General Information

Timer Settings

Design Timing Summary

Methodology Summary

Check Timing (0)

Intra-Clock Paths

Inter-Clock Paths

Other Path Groups

User Ignored Paths

Unconstrained Paths

NONE to NONE

Setup (10)

Hold (10)

Name

Slack

Levels

High Fanout

From

To

Total Delay

Logic Delay

Net Delay

Requirement

Source Clock

Destination Clock

Exception

Clock Un

Path 1

∞

11

34

a[2]

y[14]

14.510

6.213

8.298

∞

input port clock

Path 2

∞

11

34

a[2]

y[13]

14.390

6.309

8.082

∞

input port clock

Path 3

∞

10

34

a[2]

y[11]

14.236

5.974

8.262

∞

input port clock

Path 4

∞

11

34

a[2]

y[12]

14.066

6.198

7.868

∞

input port clock

Path 5

∞

11

34

a[2]

y[15]

13.937

6.301

7.636

∞

input port clock

Path 6

∞

10

34

a[2]

y[10]

13.889

6.106

7.783

∞

input port clock

Path 7

∞

10

37

a[6]

y[9]

13.686

5.717

7.969

∞

input port clock

Path 8

∞

10

37

a[6]

y[8]

12.788

5.557

7.231

∞

input port clock

Path 9

∞

9

37

a[6]

y[7]

12.414

5.234

7.180

∞

input port clock

Path 10

∞

7

33

a[0]

y[6]

11.800

5.242

6.558

∞

input port clock

Timing Summary - impl_1 (saved)

Area:

Site Type	Used	Available	Util%
Slice LUTs*	99	20800	0.48
LUT as Logic	99	20800	0.48
Bonded IOB	32	106	30.19

Final Comparison:

Metric	Base Paper Dadda	Proposed Dadda	Squarer	Integrated Base Paper Dadda and Squarer	Integrated Proposed Dadda and Squarer
Power	10.561 W	8.142 W	13.081 W	11.186 W	8.906
Timing	2.558(Hold), 12.738(Setup)	2.336(Hold), 9.977(Setup)	2.741(Hold), 12.499(Setup)	2.544(Hold), 17.372(Setup)	2.454(Hold), 13.937(Setup)
Area	0.34%	0.20%	0.22%	0.82%	0.48%
Accuracy*	80%	70%	100%	90%	85%

*Accuracy here is an approximate comparison for visualisation based on simulation results.